

METABOLIC PROFILING OF FILAMENTOUS BACTERIA ISOLATED FROM ACTIVATED SLUDGE

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Abstract

The activated sludge process is the most used biological wastewater treatment method in the world. Ever since the inception of the activated sludge process, several kinds of sedimentation problems have been observed; particularly filamentous bulking foaming which considerably compromise the efficiency of the process. Filamentous bulking has long been attributable to the extensive and uncontrollable growth of different types of filamentous bacteria. Several strategies to prevent bulking sludge have been applied but still fail as long-term control measures due to a lack of understanding of the biochemistry and physiology of the causative organisms. Effective means to long-term bulking control strategies rely on an intense understanding of filamentous bacteria in pure culture, on the basis of their biochemistry and physiology in view of corroborating these findings in situ. It was the aim of this study therefore to extrapolate on the biochemistry and physiology of previously isolated filamentous bacteria in pure culture particularly focusing on growth parameters like pH, temperature and salt tolerance, polymer hydrolyses and carbon source utilization profiles. Fifty two isolates from previous studies were investigated, however, herein are results for eight isolates. An affinity for a variety of substrates, some simple and some more complex, was displayed by the isolates. Some of the more complex substrates included starch, dextrin, peptides and lipids. This presented them with an added survival advantage over ordinary bacteria that are not as efficient with regards to hydrolyzing larger and slowly biodegradable substrates. They also preferred a variety of amino acids and sugars. The above information will add substantial value in understanding their substrate requirements and the contribution of filamentous bacteria to bulking conditions.

INTRODUCTION

The activated sludge process has been for decades now, the most used biological wastewater treatment method around the globe whose performance relies mostly on a good solid-liquid separation between the treated water and the sludge in the final clarifier [9]. Ever since the inception of the activated sludge process, filamentous bulking and foaming have been major problems affecting process efficiency. It has since been believed to date, that, the extensive and uncontrolled growth of different types of filamentous bacteria is the primary cause thereof [11]. Filamentous bacteria responsible for bulking and foaming include *Nocardia* spp., *Microthrix parvicella*, *Sphaerotilus natans*, *Haliscomenobacter hydrossis*, *Thiothrix* spp. and many more [14]. Several strategies to cure bulking and foaming which include precipitation, the use of inhibiting chemicals and the implementation of biological selectors have been applied. However, these applications are only effective as interim measures as they fail to take into account the physiology and biochemistry of these problematic filamentous bacteria and only cure the symptoms of bulking and foaming and not the cause [16]. Effective methods against bulking sludge and

foaming are therefore still urgently required. Limited information is available on physiological and biochemical characteristics of problematic bacteria [14]. Isolation of filamentous bacteria and studying them in pure culture still remains fundamental as an indispensable platform to gaining a more comprehensive knowledge of their biochemical and physiological requirements, in view of corroborating these findings *in situ* so as to develop new strategies for long-term bulking control measures. It was accordingly, the aim of this research to extrapolate on the biochemistry and physiology of pure cultures of filamentous bacteria isolated in a former study with particular focus on growth parameters such as pH and temperature, polymer hydrolyses and varying carbon source utilization attributes.

MATERIALS AND METHODS

Isolation and tentative identification of pure cultures of filamentous bacteria

The monocultural isolates of filamentous bacteria used to conduct this research were isolated, cultivated and purified according to Ramothokang *et al.* ^[16] in a previous study. A Gram stain was conducted to verify purity of isolates and also as a characterization tool. The tentative identification (Table 1) of the isolates was also as per Ramothokang *et al.* ^[16] based on the cultures' cellular characteristics using conventional microscopic techniques [2] (Results not shown).

Evaluation of physiological factors on the growth of filamentous bacteria

The effect of varying temperature (°C) and pH ranges on the growth of the isolates was evaluated. The monocultures were grown on CGYA plates for temperature evaluation and in CGYA broths for pH evaluation. The temperatures that were investigated are 10, 20 and 30 °C respectively whereas the pH ranges that are 4.5, 7.0 and 8.0. Most filamentous bacteria are reported to grow at temperatures between 20°C and 30°C [8] however, an *S. natans* isolate that was isolated from a pulp and paper industry was reported to grow well at 40°C [17] whereas there have also been reports on the ability of *M. parvicella* to grow at temperatures as low as 5 and 12°C [15]. According to Gray ^[10] bacteria require basal salt supplements such as Na, Ca, Mg and Cl for laboratory growth studies. The salt tolerance of filamentous bacteria was determined using nutrient agar at NaCl concentrations of 5%, 10%, 15% and 20%.

Evaluation of polymer hydrolyses and varying carbon source utilization attributes (Qualitative evaluation)

Polymers and varying carbon sources that were evaluated as a source of nutrients for the growth of filamentous bacteria are as follows: polymers included casein, starch, gelatin (supplemented as to growth media as the sole source of carbon), α -cyclodextrin, β -cyclodextrin, dextrin, glycogen, inulin, mannan, Tween 40 and Tween 80; lipids as edible oils included olive oil, peanut oil, sunflower oil, sesame seed oil and grapeseed oil (supplemented as to growth media as the sole source of carbon). Aromatic compound tested included vanillic acid, 2, 5- dihydroxybutyric acid (DBA), phthallic acid and o- cresol. Sugars and their derivatives, organic acids, peptides, nucleosides amino acids and many other carbon sources were evaluated. Many of the substrates were evaluated with the aid of the Biolog microplates whereas others were supplemented into growth media as the sole source of carbon. A control in each case was an inoculated medium without the carbon source. Growth was at 20-25°C.

RESULTS AND DISCUSSION

Following are tabulations of the results in terms of physiology and nutrient requirements. All isolates were Gram positive.

Table1: Physiological attributes and NaCl tolerance of pure cultures of filamentous bacteria

Temperature (°C)			pH value			NaCl tolerance (%)						
Isolate No.	10	20	30	4.5	7.0	8.0	5	10	15	20	Tentative ID	
1	+++	+++	++	++	+++	++	+	-	-	-	-	
2	++	+++	++	+++	+++	+++	+	-	-	-	-	
3	+++	+++	+++	+++	+++	+++	-	-	-	-	Thiothrix I Type 1851	
4	-	+++	+++	-	+	+++	++	-	-	-		
5	-	+++	+++	-	++	+++	++	-	-	-	-	
6	++	+++	+++	+	+	+++	+++	-	-	-	H. hydrossis	
7	++	+++	+++	+	+	+	++	-	-	-		
8	+	+++	++	+	+	++	-	-	-	-		

-: no growth +: little growth ++: moderate growth +++: excellent growth

Temperature and pH are very important parameters in the growth of all living organisms. For biochemical reactions to occur efficiently, enzymes are required that specifically perform functions that are unique to them. For these enzymes to perform at their optimum, temperature and pH need to be optimum, as enzymes are sensitive to environmental changes. Enzymes are found in all living systems and are responsible for catabolism (breakdown of substrates) and anabolism (building up of substrates). The rate at which all biochemical reactions will take place depends largely on, amongst others, the temperature and pH of the reaction medium. In general, it may be said that successful metabolism is dependent on temperature and pH. Bond *et al.* ^[12] suggested that pH is an important factor in determining whether Phosphate Accumulating Organisms (PAOs) or Glycogen Accumulating Organisms (GAOs) are favoured in EBPR systems. The filamentous type *S. natans* is known to dominate at low pH values and under acidic conditions and tends to form slimy fronds like filamentous fungi under acidic conditions [10]. Temperature has a major effect on wastewater microorganisms not only in governing the rate of reaction, but changes in the temperature also give rise to significant alterations in the community structure [10]. Some filamentous bacteria have a wide temperature tolerance ranging from 4°C to 40°C with optimal growth between 25°C to 30°C. *S. natans* has the ability to out-compete other bacteria for available nutrients at lower temperatures [10]. Seasonal conditions are also an additional factor for growth of some filamentous bacteria e.g. Nocardioforms appear to be favoured by warmer temperatures (summer) and are more likely to cause foam then [14].

With reference to Table 1, all isolates grew excellently at 20°C with moderate to excellent growth observed at 30°C with 5 of the 8 isolates growing excellently and 3 showing moderate growth. Isolates 4 and 5 did not grow at 10°C at all. At 10°C isolates 1 and 3 grew excellently and moderate growth was observed for isolates 2, 6 and 7. Little growth was observed for isolate 8 at 10°C. It may be deduced from these results that isolate 1 prefers growth at lower temperatures (excellent growth observed at 10 and 20°C) and isolates 3, 4, 5, 6, and 7 grow better at warmer temperatures (20 and 30°C). Five (isolates 2, 3, 4, 5 and 6) of the 8 isolates displayed excellent growth at pH 8.0. Isolates 2 and 3 however, seemed to grow excellently in all tested pHs i.e. pH 4.5, 7.0 and 8.0. Little

growth was observed for isolates 6, 7 and 8 at pH 4.5 and pH 7.0 but at pH 8.0 they displayed moderate to excellent growth. In an overall, it may be deduced that the isolates grow well in a wide range of temperature and pH. This may then equip these filamentous isolates with added survival advantages when growth conditions are unfavourable therefore leading to their proliferation when other bacteria fail to cope. The majority of the isolates had a salt (as NaCl) tolerance of 5% and were inhibited in terms of growth at 10, 15 and 20% salt concentrations.

Table 2: Hydrolyses of polymers

Isolate No.	Casein	Starch	Gelatin	Olive oil	Peanut oil	Sunflower oil	Grapeseed oil	Sesame oil	A-cyclodextrin	B-cyclodextrin	Dextrin	Glycogen	Inulin	mannan	Tween 40	Tween 80
1	+	-	-	+	+	+	-	-	+	+	+	-	-	-	+	+
2	+	+	+	-	+	+	-	-	+	+	+	+	+	+	+	+
3	+	-	+	-	-	+	-	+	-	-	-	+	-	+	+	+
4	+	-	-	-	+	-	-	-	+	+	+	+	+	+	+	+
5	+	+	-	-	+	+	-	+	+	+	+	+	+	+	-	-
6	-	+	-	-	-	-	-	+	-	-	-	-	-	-	-	-
7	-	+	-	-	-	-	-	+	+	+	+	-	-	-	+	+
8	-	+	-	-	+	+	+	-	-	-	+	+	-	-	+	+

Hydrolysis of starch leads to production of readily degradable monomers. The enzyme amylase is responsible for starch hydrolysis. Starch and its derivatives are the major components of biological oxygen demand (BOD) in wastewaters from industries such as textile, paper, fermentation, beverage and food processing. The high strength wastewaters can be treated by conventional active sludge processes, but at a high cost for energy consumption and sludge disposal [6]. Extensive studies have been conducted in conversion of the organic wastes to valuable products such as methane and ethanol to reduce the treatment cost and sludge disposal. Methane formation involves formation of volatile fatty acids such as acetic, propionic and butyric acids from carbohydrates by acidogenic bacteria. The organic acids would be valuable products if the low concentration acids could be recovered from the wastewater in a cost effective way [6]. Isolates were therefore tested for their ability to hydrolyze starch and only 3 out of 8 isolates gave a negative reaction (Table 2). Their ability to hydrolyze starch makes available then, the precursor substrates required for the production of organic acids which are pre-requisites for processes such as phosphorus removal in biological nutrient removal systems. Utilization of other polymeric molecules tested (α - cyclodextrin, β - cyclodextrin, dextrin, glycogen, inulin, mannan, Tween 40 and Tween 80) was also widespread. It is therefore possible that filamentous bacteria have an affinity for larger and more slowly biodegradable compounds, such as starches and lipids. This would infer a strong selective advantage over other floc-forming organisms in systems with high concentrations of such compounds, thus leading to potential bulking and foaming problems. The ability to hydrolyze milk protein (casein) was seen in all but 3 of the isolates. This again substantiates the general ability of filamentous bacteria to hydrolyze larger slowly biodegradable compounds. Systems treating dairy waste may be prone to bulking from such isolates.

Gelatin is a protein derived from collagen. The exoenzyme gelatinase first hydrolyzes gelatin into polypeptides, and then further break down the polypeptides into smaller amino acid molecules. These can then be easily transported into the bacterial cells for metabolism. While gelatin hydrolysis was not widespread with only 2 of the isolates testing positive for the hydrolysis thereof, specific populations may be present in selected systems, such as those treating abattoir wastes.

The edible oil industry has a very high specific water intake compared to other industries in South Africa and as much as 40% of this water is released as effluent into the municipal sewer systems [7]. Treatment of edible oil effluent preceding discharge into the sewers is imperative as such effluents contain a mixture of high concentrations of pollutants such as fats, oils, sodium, phosphorus, sulphates and many others. Efficient treatment of edible oil effluent is usually achieved via a combination of physical, chemical and biological treatment processes but, biological treatment of edible oil is seen as a cost- effective and efficient alternative [7].

The composition of the substrate in the influent wastewater is accepted as one of the parameters that determine the proliferation of filamentous types in activated sludge process [4]. Filaments such as *M. parvicella* have been reported to be specialized lipid consumers and consume long chain fatty acids (LCFA) [4]. Recent studies also showed that *M. parvicella* preferred long chain fatty acids especially oleic acid (major LCFA in sunflower oil) as compared to simpler substrates like glucose and acetate [4]. Due to its hydrophobic nature, *M. parvicella* is also known to play a role in foaming.

The isolates under study for this research have proven to have a considerably high affinity for some of the different edible oils tested especially, peanut oils, sunflower oil and sesame seed oil (Table 2). Essentially, while lipids are often hydrolyzed to glycerol and fatty acids, these isolates hydrolyzed the lipids (where they tested positive for lipid hydrolyses) completely such that not even fatty acids remained to react with the indicator dye used in the test media. It is therefore another notable indication that some of the filamentous bacteria existing in activated sludge processes are much more complex than it is currently known in terms of their metabolic patterns.

Table 3: Aromatic compounds, nucleosides and amino acids as C-source

C- source	1	2	3	4	5	6	7	8
Vanillic acid	+	+	+	-	-	-	-	-
2,5 DBA	+	+	-	+	+	+	+	+
Phthalic acid	+	+	-	-	-	+	+	+
O' cresol	+	+	-	+	+	+	+	+
Adenosine	+	+	+	-	+	+	+	+
2'- Deoxy Adenosine	+	+	+	-	-	+	+	+
Inosine	+	+	+	+	+	+	+	+
Thymine	+	+	-	-	-	+	+	+
Uridine	+	+	+	-	-	+	-	+
Adenosine-5'- Monophosphate	+	+	+	-	-	+	-	+
Thymine-5'- Monophosphate	+	+	-	-	-	+	-	+
Uridine-5'- Monophosphate	+	+	-	-	+	+	-	+
L- Alaninamide	+	+	-	-	+	+	-	+
D- Alanine	+	+	-	-	+	-	-	+
L- Asparagine	+	+	+	-	+	-	-	+
L- Glutamic acid	+	+	-	-	+	-	-	+
L- Serine	+	+	-	-	+	-	+	+

A lot of microorganisms have been reported not to grow in an environment with aromatic compounds [3] as they may be toxic. However, these isolates grew well on all the aromatic compounds tested. Most of the filamentous bacteria investigated presented the ability to utilize 2, 5 DBA and O'cresol, followed by Phthalic acid and then Vanillic acid (Table 3). This is a noteworthy finding in that activated sludge systems are constantly seeking means of acclimatizing their organisms such that bioremediation of the specific wastewaters is achieved. Aromatic compounds being xenobiotics, it is essential that they are removed from wastewater such that the water leaving the system is free of such compounds, especially because they are also considered as endocrine disrupting chemicals (EDCs)

posing hazards to human and animal life. The ability of bacteria to catabolise nucleosides has been shown with organisms such as *Bacillus subtilis* [13]. Nucleosides phosphorylases are involved in the intracellular metabolism of nucleosides obtained from the external environment or arising from intracellular breakdown of nucleotides [13]. It is demonstrated in Table 3 that the nucleosides under study (2'- Deoxy Adenosine, Inosine, Thymine, Uridine, Adenosine-5'- Monophosphate, Thymine-5'- Monophosphate and Uridine- 5'- Monophosphate) were generally widely utilized by the isolates although they tested negative for some. Inosine was utilized by all 8 isolates. Amino acids are the building blocks of proteins. According to Ubukata and Takii ^[19], there are a limited number of microbial populations that can assimilate amino acids in the activated sludge system. As seen in Table 3, only isolate 4 was negative for all the amino acids whereas isolates 1, 2, 5 and 8 were positive for all tested amino acids. It may, based on this study, be generally accepted that these filamentous bacteria have the capacity to use and assimilate amino acids in their external environment for metabolism. This would then present them with an advantage in terms of cellular regeneration when conditions in the wastewater system become limiting for bacterial growth.

Table 4: Organic acids as C-source

C-Source	1	2	3	4	5	6	7	8
Acetic acid	+	+	-	-	+	-	+	+
α - Hydroxybutyric acid	-	+	-	-	+	-	-	+
β - Hydroxybutyric acid	+	+	-	-	+	+	-	+
γ - Hydroxybutyric acid	-	+	-	-	+	-	+	-
p- Hydroxy- Phenylacetic acid	+	+	+	-	+	-	-	-
α - Ketoglutaric acid	+	+	-	-	+	-	+	+
α - Ketovaleric acid	+	+	+	-	+	-	-	+
L- Lactic acid	+	+	-	-	+	-	-	+
D- Malic acid	+	+	+	-	+	-	-	+
L- Malic acid	+	+	-	-	+	-	-	+
Propionic acid	+	+	-	-	+	-	-	+
Pyruvic acid	+	+	+	-	+	-	+	+
D- Lactic Acid Methyl Ester	-	+	-	-	-	-	-	-
Pyruvic Acid Methyl Ester	+	+	-	-	+	-	-	+
N- Acetyl- L- Glutamic Acid	+	+	-	-	-	-	-	-

Microorganisms in the activated sludge utilize organic compounds for both energy and as a carbon source. And, for efficient P-removal in biological nutrient removal systems, volatile fatty (VF)/ organic acids are a pre-requisite. Essentially, bacteria degrade organic acids and use them as a carbon and energy source for P- removal later on. Yu ^[6] also states that the organic acids can also be utilized and polymerized into polyhydroxyalkanoate (PHA) by bacteria whereby complex organic compounds (carbohydrates, fats, and proteins) are converted to simple organic compounds, mainly short-chain volatile organic acids (acetic, propionic, and lactic acids). It is in view of this information therefore, that it is noteworthy to point that the majority of isolates under study utilized, on average, all of the organic acids tested (Table 4). This finding may have positive implications on P-removal and the optimization thereof. It is widely accepted that the current understanding of the microbiology of phosphorus removal in activated sludge is still incomplete; this then necessitates the need to explore in detail the role filamentous bacteria play in P-removal systems.

Table 5: Sugars and sugar derivatives as C- source

C- source	1	2	3	4	5	6	7	8
L- Arabinose	+	+	+	+	+	+	-	+
D- Cellobiose	-	+	+	+	+	-	+	-
D- Fructose	-	+	+	+	+	-	-	+
L- Fucose	-	-	+	+	+	-	-	+
D- Galactose	-	+	+	+	+	-	-	+
Gentiobiose	-	+	+	+	+	+	-	+
α - D- Glucose	+	+	+	+	+	-	-	+
m- Inositol	-	+	-	+	+	-	+	-
α - D- Lactose	-	+	-	+	+	+	-	-
Lactulose	+	+	+	+	+	+	+	-
Maltose	+	+	+	+	+	+	+	+
Maltotriose	-	+	+	+	+	+	+	+
D- Mannose	-	+	+	+	+	-	+	+
D- Melezitose	-	+	+	+	+	+	+	-
D- Melibiose	-	+	+	+	+	-	+	+
Palatinose	+	+	+	+	+	-	+	-
D- Psicose	-	+	+	+	+	+	+	+
D- Raffinose	-	+	+	+	+	-	-	-
L- Rhamnose	-	+	+	+	+	+	-	-
D- Ribose	-	+	+	+	+	-	+	+
Stachyose	-	+	+	+	+	-	-	-
Sucrose	-	-	+	+	+	-	-	-
D- Tagatose	-	+	+	+	+	+	+	-
D- Trehalose	-	+	+	+	+	-	+	+
Turanose	+	+	+	+	+	-	+	-
Xylitol	-	+	-	+	+	-	+	+
D- Xylose	+	+	+	+	+	+	-	-
3- Methyl- Glucose	-	+	-	+	+	-	+	-

Identifying the relationships between various bacterial populations and the substrates they consume is essential to the understanding of population dynamics and to the development of process control in activated sludge [1]. Sugars are some of the simplest form of carbon available to microorganisms as most of them are easier to utilize and readily biodegradable due to their simpler structure. In wastewater treatment systems, the availability of carbon to bacteria is essential for effective P-removal and N-removal (nitrogen removal). All the 8 isolates had an affinity for at least 7 of the 28 sugars and sugar derivatives tested (Isolate 1, Table 5). Isolates 4 and 5 used all 28 sugars and their sugar derivatives tested. Generally, the utilization of the substrates in Table 5 was widespread across all isolates. This further substantiates the diversity of the metabolic patterns of filamentous bacteria.

CONCLUSION

The growth of filamentous bacteria in an activated sludge is determined by whether they can obtain energy for their growth from substrates available to them and how versatile their energy generating systems are to cope with the variations on nutritional and environmental conditions possible to occur [14]. It is concluded therefore, that with their metabolic patterns being so evidently diverse, it still remains imperative to elucidate filamentous bacteria in terms of their metabolic traits as a fundamental step to gaining a better understanding of filamentous sludge bulking, foaming and biological nutrient removal and the optimization thereof.

CURRENT RESEARCH

Research is ongoing and verification of the identification of the isolates using Fluorescent *In Situ* Hybridization (FISH) is currently being finalized.

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